

Assignment: Day 1

Introduction to Networking and the OSI Model

Q1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Ans: I installed Wireshark, selected my Wi-Fi interface, started the capture, opened instagram.com in the browser, and stopped the capture after the page loaded. The Protocol column of the packet list showed these protocols:

Protocol	What it was doing in the capture
DNS	Converted the name instagram.com into an IP address
TCP	Three-way handshake (SYN, SYN-ACK, ACK) and reliable data transfer
TLSv1.2 / TLSv1.3	Encrypted the connection (Client Hello, Server Hello, application data)
QUIC / UDP	Fast encrypted transport that modern sites use instead of TCP
HTTP / HTTP2	Web requests and responses (mostly hidden inside TLS)
ARP	Found the MAC address of the router on the local network
ICMP / mDNS / SSDP	Background network messages from the operating system and other devices

Instagram uses HTTPS, so most of the traffic appeared as TLS or QUIC and the actual page content was not readable.

Q2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents.

Ans: I picked three packets from the capture and mapped each one to its OSI layer using the Protocol and Info columns.

Packet	Protocol	Info column (summary)	Main OSI layer
1	DNS	Standard query A instagram.com	Layer 7: Application
2	TLSv1.3	Client Hello	Layer 6: Presentation (encryption)
3	TCP	[SYN] Seq=0 Win=64240 (port 51234 to 443)	Layer 4: Transport

The DNS packet carries a name-lookup request made by an application. The TLS packet sets up encryption for the data. The TCP packet contains port numbers and flags used to establish a connection.

Q3. Create a colorful diagram of the OSI Model using any drawing tool. For each layer, add a real-world example from apps you use daily.

Ans:

#	Layer (Data Unit)	Function / Protocols	Real-world example
7	Application (Data)	User-facing services and protocols: HTTP, DNS, SMTP, FTP	<i>WhatsApp chat, Chrome browsing, Gmail</i>
6	Presentation (Data)	Data format, compression and encryption: TLS, JPEG, MP4	<i>Instagram photos (JPEG), encrypted chats</i>
5	Session (Data)	Starts, maintains and ends a conversation between two devices	<i>Staying logged in on Instagram, a Google Meet call session</i>
4	Transport (Segment)	End-to-end delivery using port numbers: TCP, UDP	<i>Text messages over TCP, voice calls over UDP</i>
3	Network (Packet)	Logical (IP) addressing and routing: IP, ICMP	<i>Ping to 8.8.8.8, your router forwarding data</i>
2	Data Link (Frame)	MAC addressing and delivery on the local network: Ethernet, Wi-Fi	<i>Phone connecting to the home Wi-Fi router</i>
1	Physical (Bits)	Sends raw bits as electrical, light or radio signals	<i>Wi-Fi radio waves, LAN cable, fibre broadband</i>

Q4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Ans: Suppose I type "Hello" and press send. On my phone the message moves down the layers, and on my friend's phone it moves back up.

- **Application layer (Layer 7):** I type "Hello" in WhatsApp and tap send. The app creates the message and hands it to the network services to send.
- **Presentation layer (Layer 6):** The message is converted into a proper data format and encrypted (WhatsApp uses end-to-end encryption), so nobody in between can read it. Any photo or video would also be compressed here.
- **Session layer (Layer 5):** WhatsApp keeps an open, logged-in connection with its servers. This layer manages that ongoing conversation so the connection stays alive while I chat.
- **Transport layer (Layer 4):** The data is cut into small segments and given port numbers. TCP numbers each segment, checks that all of them arrive, and resends any that get lost.
- **Network layer (Layer 3):** Each segment is placed inside a packet with my IP address as the source and the server's IP address as the destination. Routers use these IP addresses to choose the path across the internet.
- **Data Link layer (Layer 2):** The packet is put into a frame with the MAC address of my Wi-Fi router as the next hop, so it can be delivered inside my local network.
- **Physical layer (Layer 1):** The frame is turned into bits and sent as radio waves from my phone to the router, then as electrical or light signals through cables and fibre.

At the receiving end the process runs in reverse. My friend's phone receives the bits, rebuilds the frames, packets and segments, decrypts the data, and the WhatsApp app finally shows "Hello" on the screen.